

Applied Research Project Proposal

*Developing a Diabetes and Prediabetes Prediction Web App
Using 2025 CDC BRFSS Data*

CSIS 4495 002 Applied Research Project

Student Name: Liqin Shi

Student ID: 300431612

Project arrangement: Individual project

Submission date: September 28, 2026

1 Introduction and problem definition

Diabetes and prediabetes are major public health concerns. The CDC National Diabetes Statistics Report estimates 40.1 million people in the United States had diagnosed or undiagnosed diabetes in 2023 and 115.2 million adults had prediabetes. These are population estimates from sources other than the 2025 BRFSS file and are provided as background, not as findings of this project ^[1].

This project asks whether information that adults can report themselves, such as age, body mass index, physical activity, smoking and selected health history, can distinguish three reported statuses: no reported diabetes or prediabetes, reported prediabetes, and reported diabetes. The output will be a research demonstration and educational web app. It will not diagnose diabetes, estimate a person's future incidence from longitudinal follow-up, or replace laboratory testing.

1.1 Questions and project goals

The main question is whether I can turn the 2025 CDC survey data into a working web app that predicts reported diabetes status. I will ask:

- 1) Which survey fields are suitable for a short web form?
- 2) Which of the two models classifies the three groups more accurately on test data?
- 3) Can the app present its results clearly to users? Both models will predict the same three groups: no diabetes, prediabetes and diabetes.

I will begin with a few survey fields that people can answer in a short form. The main goal is a working web app, with clear data preparation and an honest assessment of how well the three categories are predicted.

1.2 Prior work and research gap

The CDC BRFSS is a large public survey of adult health conditions and behaviours ^[2]. The National Diabetes Statistics Report shows why distinguishing diabetes and prediabetes matters, while also showing that laboratory-measured prediabetes is different from being told about prediabetes ^[1]. This project will use the survey to build a product that predicts reported status. The project will show how CDC data can be prepared, used to train a machine learning model, and presented in a web app. A few Power BI visuals will help describe the data.

2 Data source, sampling and outcome definition

The source is the CDC 2025 Behavioral Risk Factor Surveillance System combined landline and cellular telephone public-use file (LLCP2025 XPT), downloaded from the official CDC page ^[2]. No participants will be recruited and no personal clinical records will be collected. The 2025 combined file represents participating jurisdictions, not every US jurisdiction: California, Mississippi, Nevada and the US Virgin Islands did not meet the inclusion requirements ^[2].

I have downloaded and opened the CDC file in a Jupyter notebook. My working data frame has 356,158 rows and 283 columns. I checked the column names and missing values; 108 columns have more than 80% null values. The CDC page lists 284 variables in the original XPT file, so I will check this difference while preparing the data. Next, I will use the codebook to choose a short list of inputs for the app.

2.1 Outcome and available sample

I plan to use DIABETE4 to define three groups: code 3 for no diabetes, code 4 for prediabetes or borderline diabetes, and code 1 for diabetes. I will verify these meanings against the 2025 codebook and count valid responses before creating the target variable [3]. The optional PREDIAB2 module is not needed for these groups.

I will check the response wording and special codes in the 2025 codebook. I will exclude pregnancy-only diabetes and unknown or refused responses. I will count the records in each group and report the model's performance for all three groups.

2.2 Data preparation for the product

I will start with a short list of survey questions, such as age group, body mass index, physical activity and selected health history. I will confirm the available fields and missing values before choosing the final app inputs. Questions that directly reveal the diabetes answer will not be used as inputs.

Power BI will show the three groups and a few simple comparisons, such as age group or body mass index. Charts will clearly distinguish survey record counts from any weighted estimates.

3 Methods and implementation

I will check the codebook, define the three labels, clean the selected data, and split it into training and test sets. Then I will train logistic regression and random forest, compare their results and connect a saved model to the Flask app. I will document the main steps in a notebook.

I will evaluate the models on test data using a confusion matrix and F1 score for each group. I will pay particular attention to prediabetes, which was difficult to predict in my earlier project. I will also check a few example inputs in the app.

3.1 Technologies and architecture

Layer	Proposed technology and function
Operating system/platform	Windows 11 for development; local browser deployment for demonstration.
Analysis and models	Python, pandas, scikit-learn and Jupyter for cleaning, model training and evaluation.
Visualization	A few Power BI visuals showing class counts and selected comparisons.
Backend	Flask loads the saved model and accepts answers from the web form.
Frontend	HTML and CSS for a short form and an easy-to-read result page.
Database	SQLite stores a small table of cleaned summary data for the app to display. The saved model makes the predictions.
Version control	GitHub repository with notebooks, source, documentation and reproducible setup instructions.

The app will use the saved model to display a predicted category and a short explanation of what the result means. If the probabilities are useful and understandable, I may also display them. The app will not ask for names or other direct identifiers.

4 Expected results, validity and utility

The final product will include a data preparation and modelling notebook, a saved model, a working Flask app, a few Power BI visuals, a simple SQLite table, a README and a final report. The user will enter answers in the web form and see a predicted category with a brief explanation. The app will say that it is a course demonstration and not a diagnostic tool.

I will compare logistic regression and random forest on all three groups, with particular attention to prediabetes. I will choose the model for the app after reviewing the results and describe where it performs well or poorly.

I will check that the form accepts valid answers, handles missing answers, and shows a result from the saved model. I will also compare the Power BI counts with the notebook and check that the README explains how to run the app.

The main limitations are self-reported answers, missing responses and the fact that the 2025 survey is a snapshot. The app is a course demonstration and cannot determine whether a person has diabetes.

5 Project management and milestones

Period	Milestone	Evidence or deliverable
Sep 21–27	Select topic; examine source and template; inspect raw file; draft proposal.	Proposal PDF; repository structure; initial notebook.
Sep 28–Oct 11	Codebook review; reconcile 283/284 variables; verify DIABETE4 codes and class counts.	Three-class count table; data dictionary; documented label definition.
Oct 12–25	Clean data, EDA and survey summaries.	Cleaning notebook; cleaned summaries; Power BI draft and missingness figures.
Oct 26–Nov 8	Train and compare two models.	Training notebook and model comparison.
Nov 9–22	Evaluate results and connect the Flask app to the saved model and SQLite summaries.	Model results, working local app and SQLite table.
Nov 23–Dec 6	Test, document and prepare final materials.	README, test scenarios, final report and demo.

These dates are targets and may change with course deadlines. As an individual student, I will manage the research, data, app and final submission.

5.1 Project management chart

Work package	Sep 21-30	Oct 1-15	Oct 16-31	Nov 1-15	Nov 16-30	Dec 1-6
Proposal and source audit	●	●				
Data preparation		●	●			
Models and evaluation			●	●		
Power BI visuals			●			
Flask app and SQLite				●	●	
Testing and report					●	●

Each phase produces a small repository commit or short written progress record. Code and documentation will be committed as work is completed, allowing the instructor to review progress rather than receiving a single final upload.

6 Project contract

I, Liqin Shi, agree to carry out the individual project within the research scope, milestones and deliverables described in this proposal. I will document the three-class label definition and any changes to the plan, maintain a dated work log, cite external sources, disclose AI assistance, and submit the required course reports and code. I will use only public survey data and will not collect identifiable health information in the demonstration app.

Student Li Shi Date: September 28, 2026

7 AI tools and specific use

AI tool, version and account	Specific use	Student contribution beyond AI
ChatGPT Codex, GPT-6, personal ChatGPT account	Helped review CDC documentation, discuss project feasibility and architecture, and edit the proposal.	Selected and inspected the CDC data; reviewed and revised the proposal. I will complete the analysis and build the app.

8 Work log for Liqin Shi

Date	Hours	Work completed
Sep 21, 2026	4	Investigated the use of medical notes; reviewed privacy concerns and decided not to use those data.
Sep 22, 2026	4	Searched for a replacement dataset; selected CDC 2025 BRFSS and framed the diabetes project.
Sep 23, 2026	4	Discussed the CDC source with the instructor; organized the GitHub project and uploaded initial materials.
Sep 24, 2026	2	Studied relevant data documentation and considered project architecture.
Sep 25, 2026	2	Continued background study and architecture planning.
Sep 27, 2026	5	Reviewed proposal requirements and CDC documentation and prepared proposal draft.
Sep 28, 2026	1	Completed a final review of the proposal.

I will continue to record the date, time spent and work completed as the project progresses.

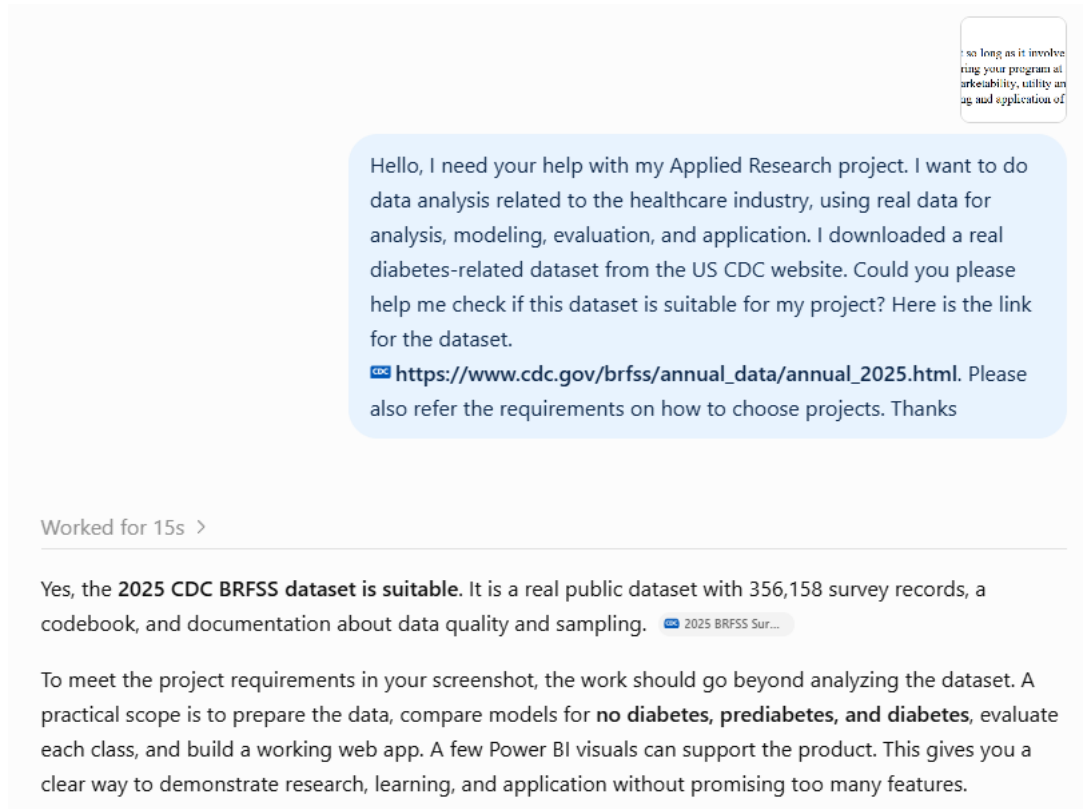
9 References

- [1] Centers for Disease Control and Prevention. National Diabetes Statistics Report, updated 2026. <https://www.cdc.gov/diabetes/php/data-research/> (accessed September 27, 2026).
- [2] Centers for Disease Control and Prevention. 2025 BRFSS Survey Data and Documentation. https://www.cdc.gov/brfss/annual_data/annual_2025.html (accessed September 27, 2026).
- [3] Centers for Disease Control and Prevention. 2025 BRFSS Codebook. Available from the 2025 BRFSS Survey Data and Documentation page: https://www.cdc.gov/brfss/annual_data/annual_2025.html (accessed September 27, 2026).
- [4] Centers for Disease Control and Prevention. Complex Sampling Weights and Preparing 2025 BRFSS Module Data for Analysis, August 2026. https://www.cdc.gov/brfss/annual_data/2025/pdf/Complex-Sampling-Weights-and-Preparing-Module-Data-for-Analysis-2025-508.pdf (accessed September 27, 2026).

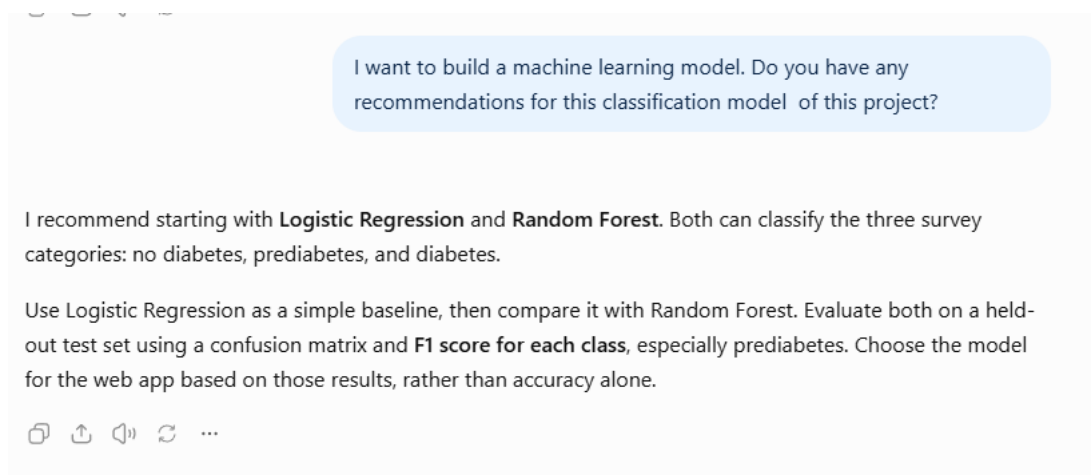
Appendix A AI prompt history

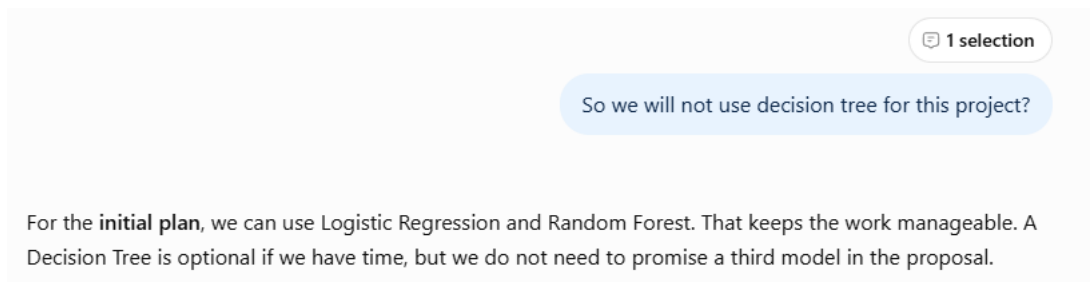
The prompts below are included as a record of AI assistance during proposal preparation.

Prompt 1, September 28: Asked whether the CDC 2025 dataset is suitable for this project. The screenshot below shows the original prompt and response.

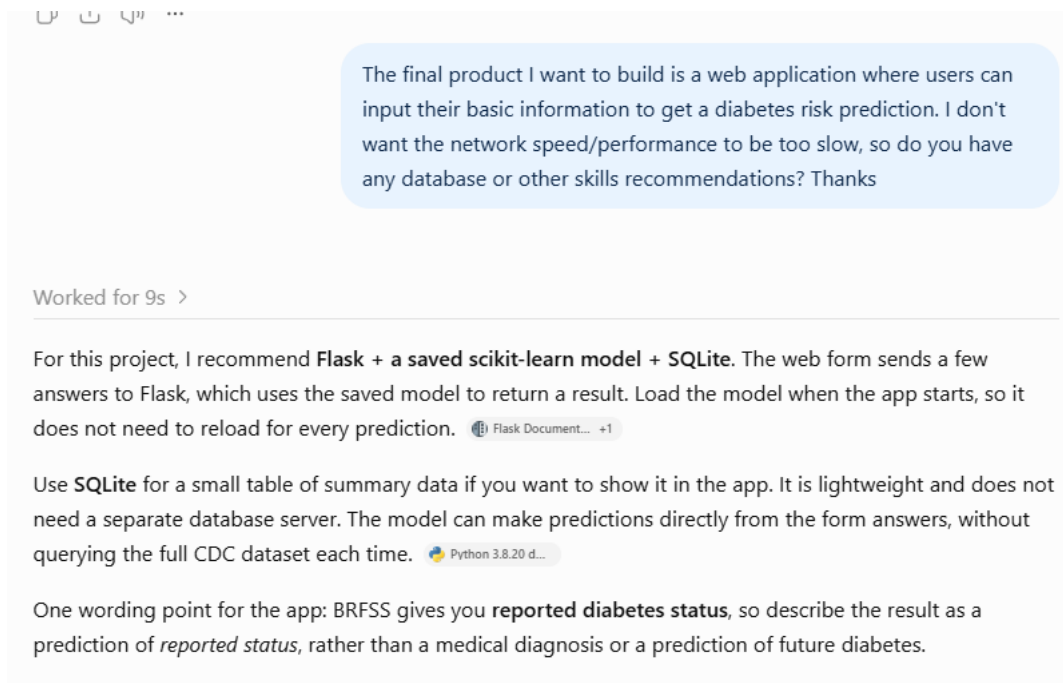


Prompt 2, September 28: Asked which classification models to use. The screenshot below shows the original prompt and response.

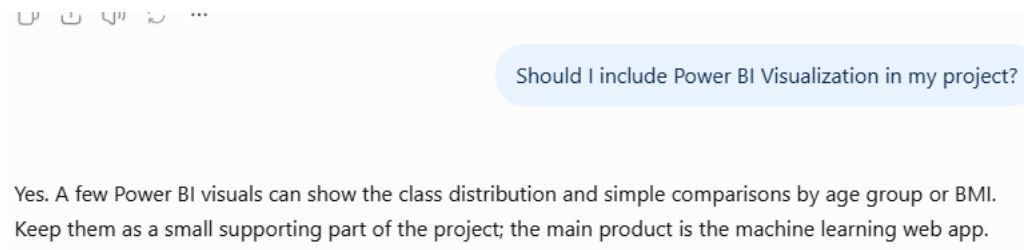




Prompt 3, September 28: Asked how to build a responsive web app and which database to use.



Prompt 4, September 27: Asked whether Power BI visualizations should be included.



Prompt 5, September 28: Requested a final review of the proposal before submission

 **LiqinS_Proposal(1).docx**
Document

I have finished the first draft. Please help me check the grammar and see if the structure of the submission needs any revision, to make it clear and concise.

Worked for 1m 55s >

I proofread the draft and made focused edits for clearer, simpler English. I also corrected the schedule header, updated the work-log note to match the hours you entered, and removed a submission reminder from the report. Your Word formatting and project scope are preserved.

Before submission: please verify the work-log hours and dates, sign the project contract, and check that Appendix A contains the prompt history your instructor requires.

 **LiqinS_Proposal_proofread.docx**
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